

Skills Summary

Finding a Percent of a Quantity

Use this reference while completing your worksheet or when studying.

Reading an Amount off a Hundredths Grid

Rule: “Percent” means *per one hundred*. A hundredths grid cuts the whole quantity into 100 equal parts, so one part is 1% of it. **one part** = whole $\div$ 100, and **p percent of the whole** = $p \times$ (one part) = $\frac{p}{100} \times$ whole.

Example: A hundredths grid stands for a jar of 300 marbles, and 40 parts are shaded. How many marbles are shaded?

Step 1. The grid always has 100 parts, so first find what one part is worth — that turns a percent into a count.

Step 2. One part is $300 \div 100 = 3$ marbles, so 40 parts are $40 \times 3 = 120$ marbles.

Try it: A hundredths grid stands for 250 stickers and 60 parts are shaded. How many stickers are shaded?

check: 150

Building a Percent from Ten-Percent Strips

Rule: Cut a tape of the whole amount into 10 equal strips. One strip is 10% (whole $\div$ 10); half a strip is 5%; a tenth of a strip is 1%. Any friendly percent is a sum of strips: $35\% = 10\% + 10\% + 10\% + 5\%$.

Example: Find 35% of 80 dollars without multiplying decimals.

Step 1. Thirty-five percent is not a single strip, so build it: three whole ten-percent strips plus one half strip.

Step 2. One strip is $80 \div 10 = 8$ dollars and half a strip is 4 dollars, so 35% is $8 + 8 + 8 + 4 = 28$ dollars.

Try it: Find 45% of 60 dollars with ten-percent strips.

check: 27

Multiplying by the Decimal Form

Rule: Divide the percent by 100 to get its decimal form — the decimal point moves two places to the left — then multiply. **p percent of q** = $(p \div 100) \times q$. This is the fastest route when the percent is not friendly, such as 7.5% or 23%.

Example: Find 24% of 150 people.

Step 1. Twenty-four percent is not a whole number of ten-percent strips, so convert to a decimal and multiply.

Step 2. $24 \div 100 = 0.24$, and $0.24 \times 150 = 36$ people.

Try it: Find 6% of 450 kilometres.

check: 27

(!) Watch out: 8% is 0.08, not 0.8. Moving the decimal point only one place turns eight percent into eighty percent, and the answer comes out ten times too big.